

Test III

Name
Date
Course

- I) Write your name, the date, and the course title (i.e. Trigonometry).
- II) Show all work.
- III) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) You will need a calculator for this test (be sure that it is in degree mode).
- V) Each problem is worth four points for a total of 100 points.

Find the values of the following expressions without a calculator.

① $\sin^{-1}(\frac{1}{2})$

② $\cot^{-1}(-1)$

Find the value of the following expression. You will need a calculator. Round to the nearest tenth.

③ $\cos^{-1}(-0.77)$

Write an equation corresponding to each graph.
★ There is more than one equation that can be written for each graph (e.g. cosine functions or sine functions can represent the same graph). To avoid confusion, use the sine or cosecant functions rather than the cosine or secant functions. ★ Each horizontal unit on the graphs is 90° . Each vertical unit is one.

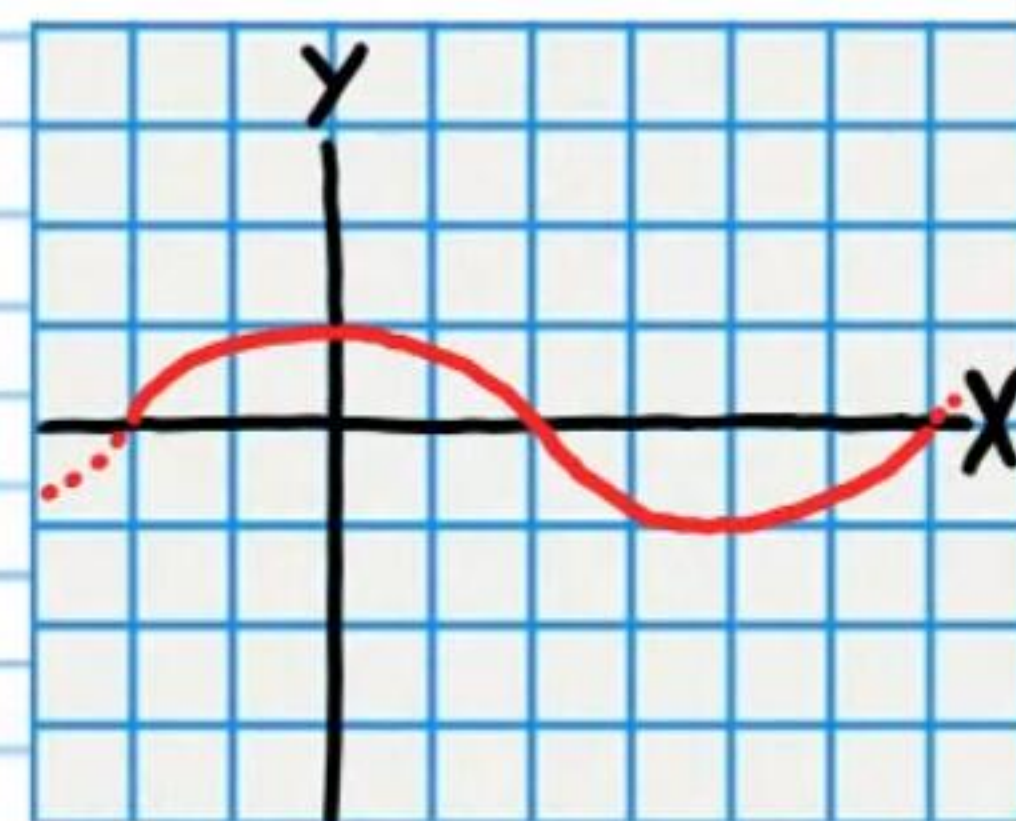
④ General Form of the Sine Function: $y = A \sin(\omega x - \phi) + B$

Midline Equation:

Amplitude:

Omega:

Phase Shift:

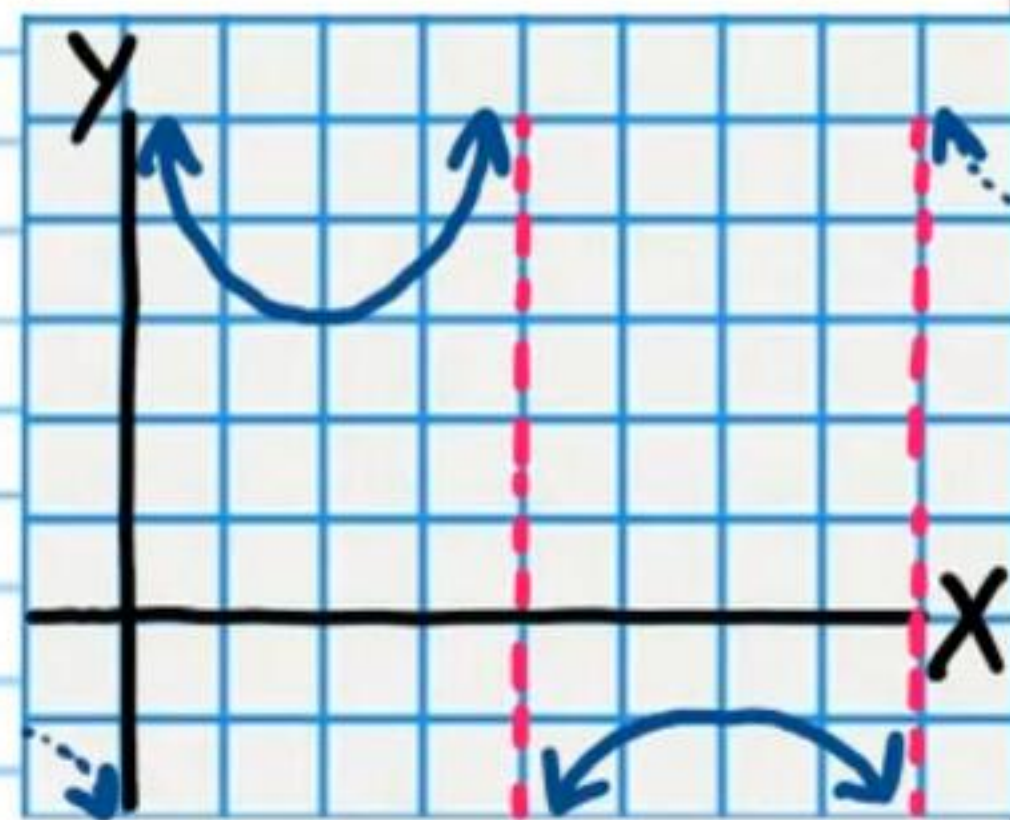


⑤ General Form of the Cosecant Function: $y = A \csc(\omega x - \phi) + B$

Midline Equation:

A:

Omega:



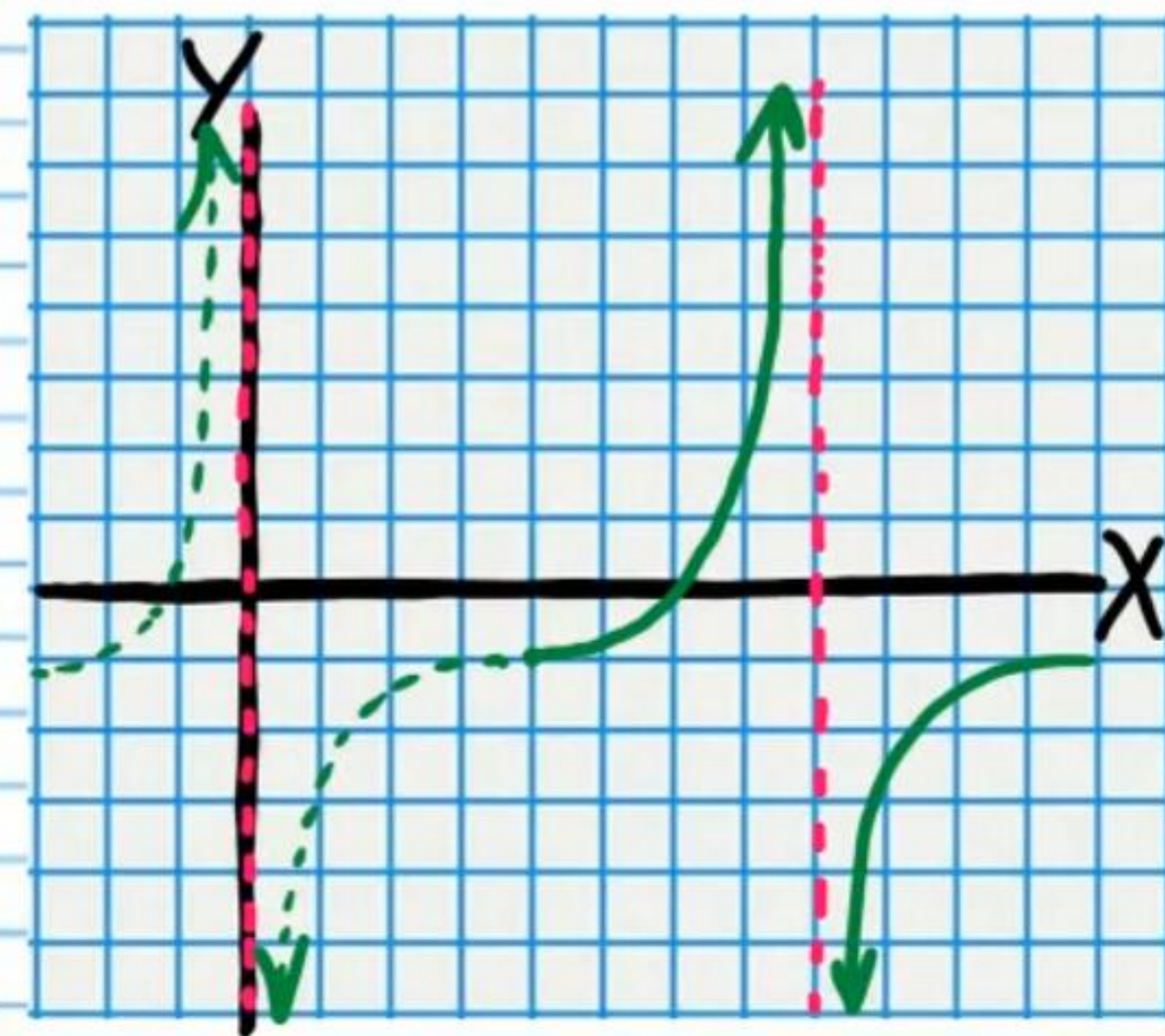
Phase Shift:

⑥ General Form of the tangent Function: $y = A \tan(\omega x - \phi) + B$

Midline Equation:

A: ★ Assume $A=1$.

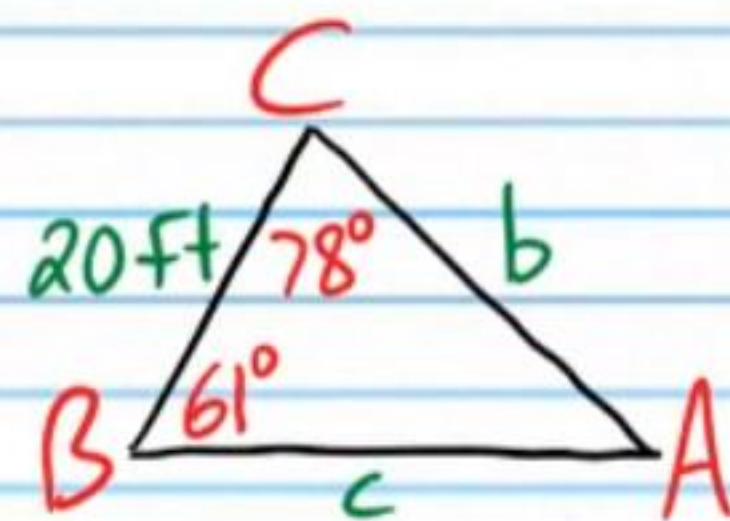
Omega:



Phase Shift:

Find the missing side lengths and angle measures of each triangle. In other words, solve each triangle. There might not be a triangle that exists for some of these problems.

⑦ $m\angle A$

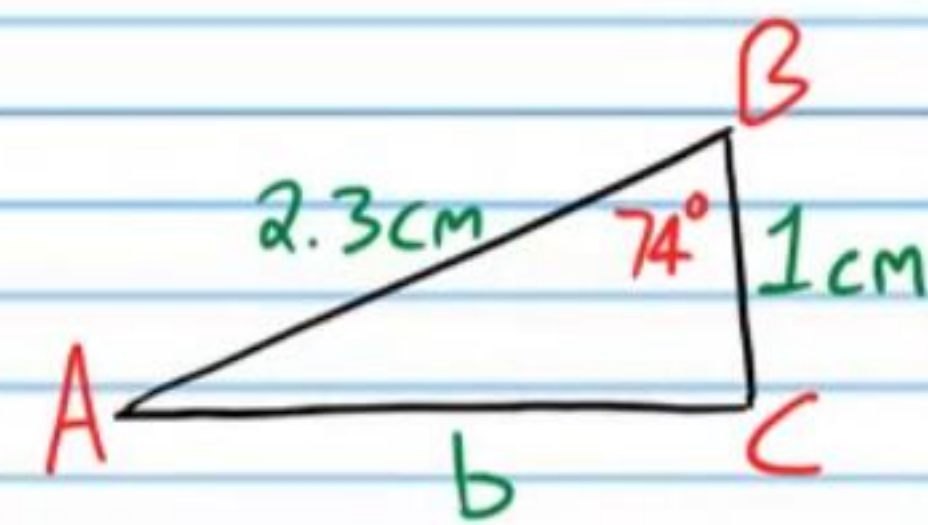


⑧ b

⑨ c

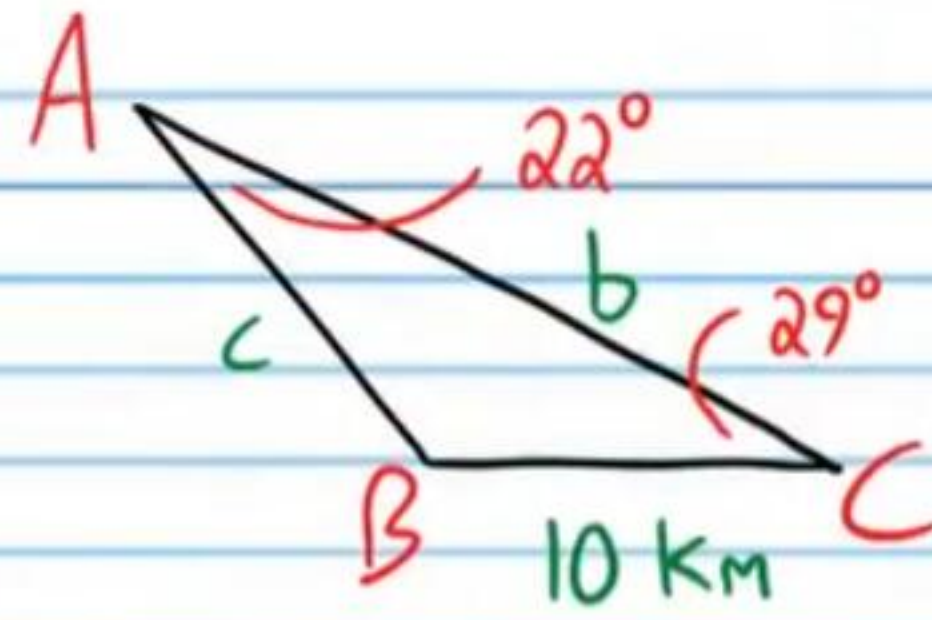
⑩ b

⑪ $m\angle A$



⑫ $m\angle C$

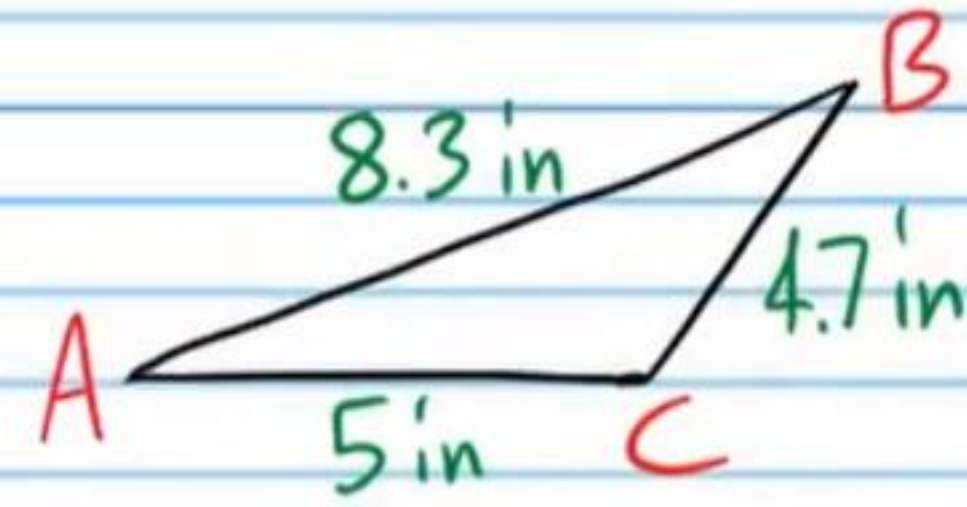
⑬ $m\angle B$



⑭ c

⑮ b

⑯ $m\angle C$



⑰ $m\angle A$

⑱ $m\angle B$

①⑨ $A = 109^\circ$, $b = 4.2 \text{ m}$, $a = 4 \text{ m}$

②② $B = 41^\circ$, $b = 31\text{ cm}$, $c = 70\text{ cm}$

② $C = 57^\circ, c = 34 \text{ mi}, b = 30 \text{ mi}$

②② $A = 28^\circ, c = 8 \text{ in}, a = 6.4 \text{ in}$

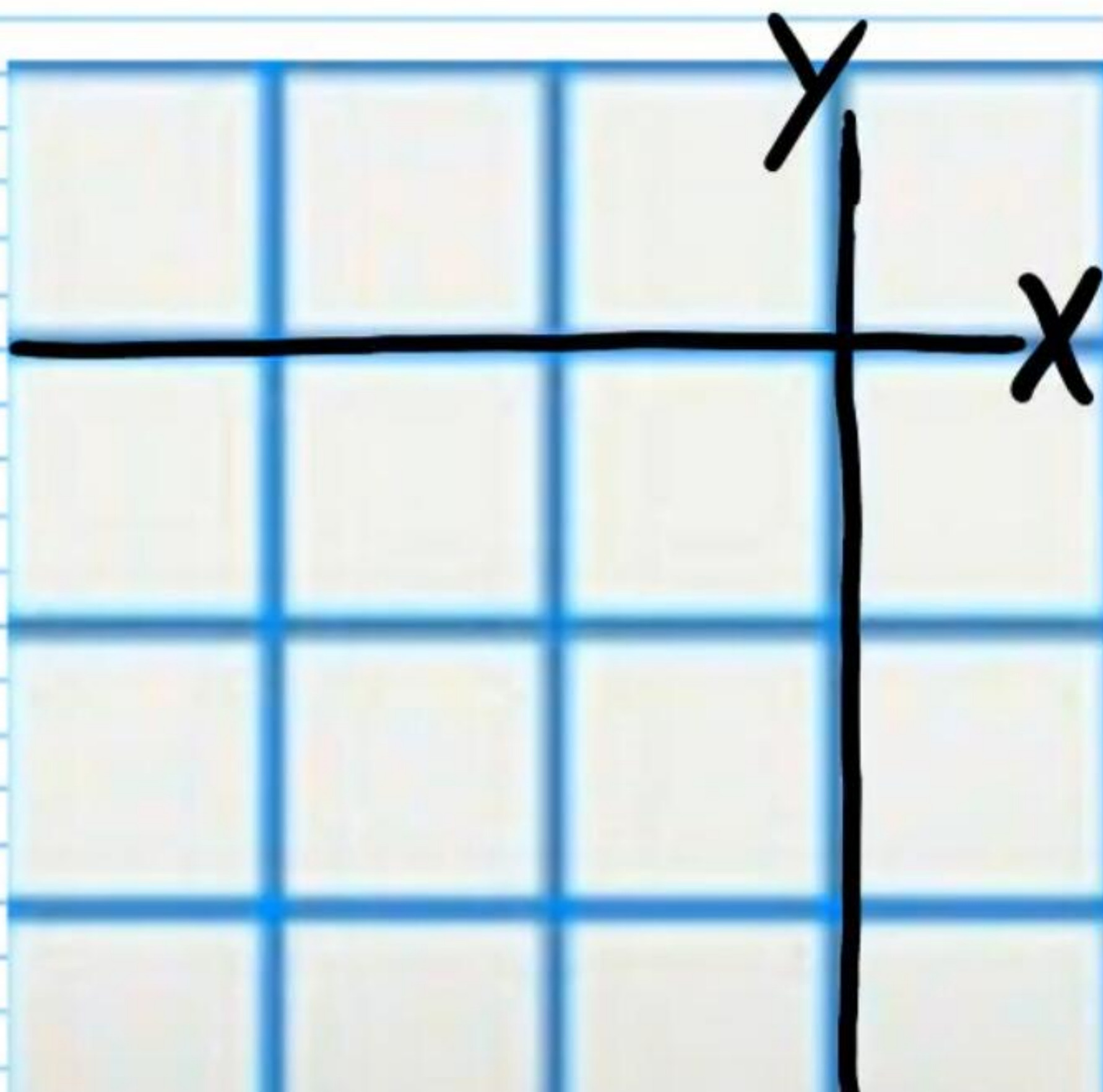
For the following problems, write the equation of the midline, and find the period, amplitude, and phase shift. Then, graph the equation. *Only one cycle of the graph is necessary.*

②③ $y = \sec(4x + 360^\circ) - 1$

Midline Equation:

Period:

Phase Shift:



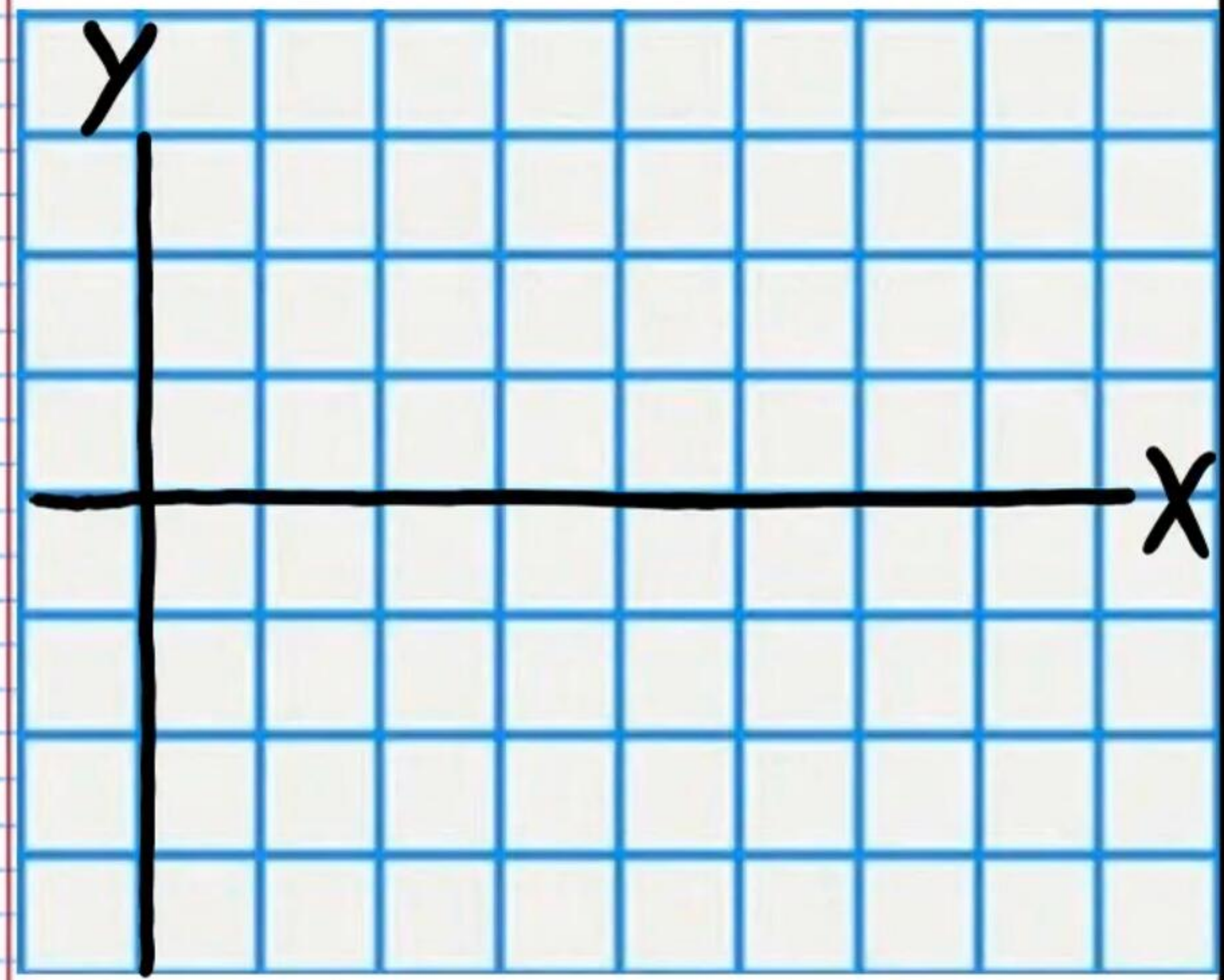
②④ $y = -2\sin(2x - 1080^\circ)$

Midline Equation:

Period:

Phase Shift:

Amplitude:

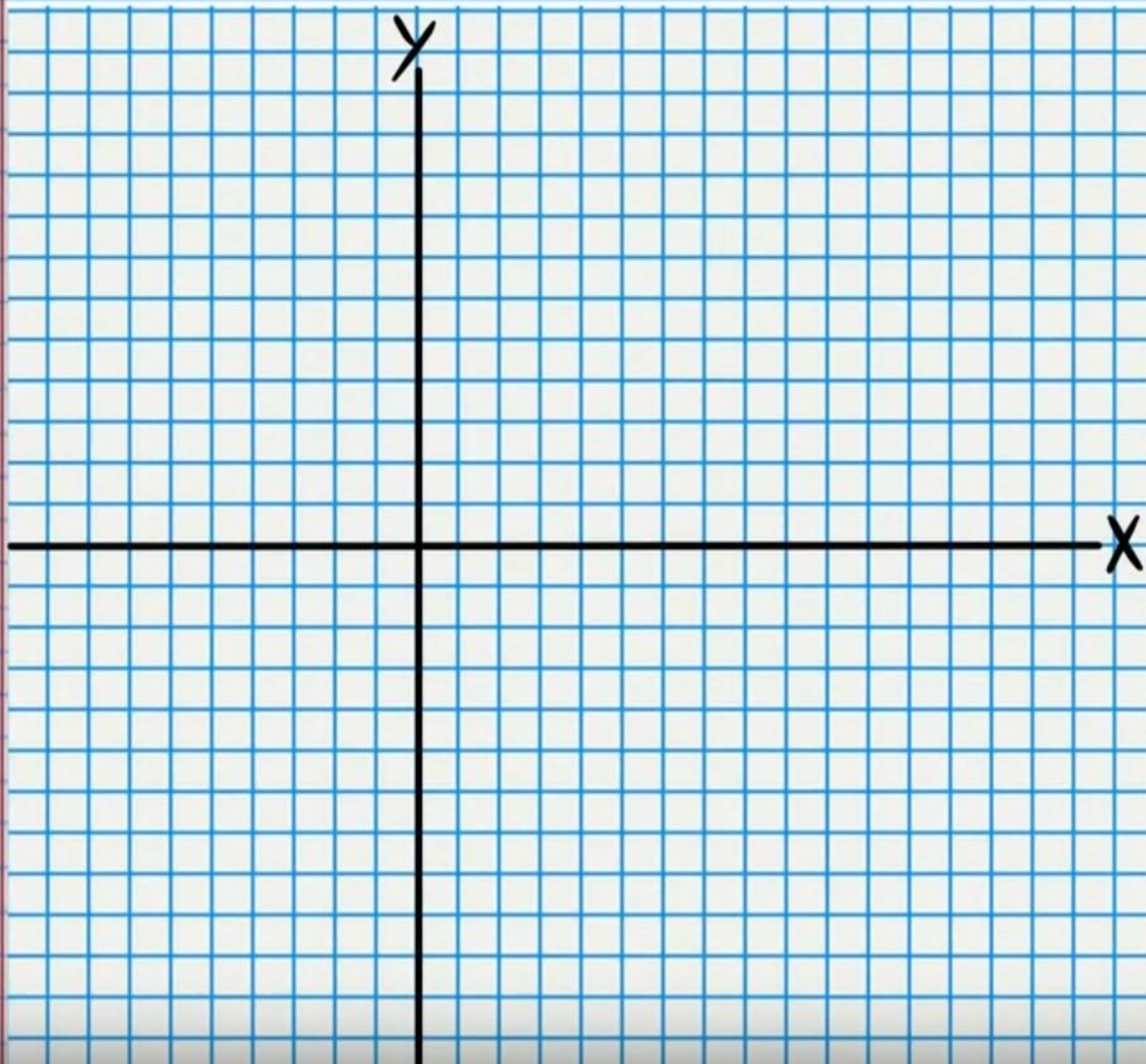


②⑤ $y = \tan\left(\frac{x}{4} + 45^\circ\right) + 1$

Midline Equation:

Period:

Phase Shift:



Test III Answers

① 30°

② 135°

③ 140.4°

④ $y = 5 \sin\left(\frac{x}{2} + 90^\circ\right)$

⑤ $y = 2 \csc\left(\frac{x}{2}\right) + 1$

⑥ $y = \tan\left(\frac{x}{4} - 90^\circ\right) - 1$

⑦ $A = 41^\circ$

⑧ $b \approx 26.66 \text{ ft}$

⑨ $c \approx 29.82 \text{ ft}$

⑩ $b \approx 2.24 \text{ cm}$

⑪ $A \approx 25.4^\circ$

⑫ $C \approx 80.6^\circ$

⑬ $B = 129^\circ$

⑭ $c \approx 12.94 \text{ km}$

⑮ $b \approx 20.75 \text{ km}$

⑯ $C \approx 117.64^\circ$

⑰ $A \approx 30.11^\circ$

⑱ $B \approx 32.26^\circ$

⑲ $A + B \neq 180^\circ$

$$109^\circ + 83.12^\circ \neq 180^\circ$$

$$192.12 \neq 180^\circ$$

No Triangle Possible

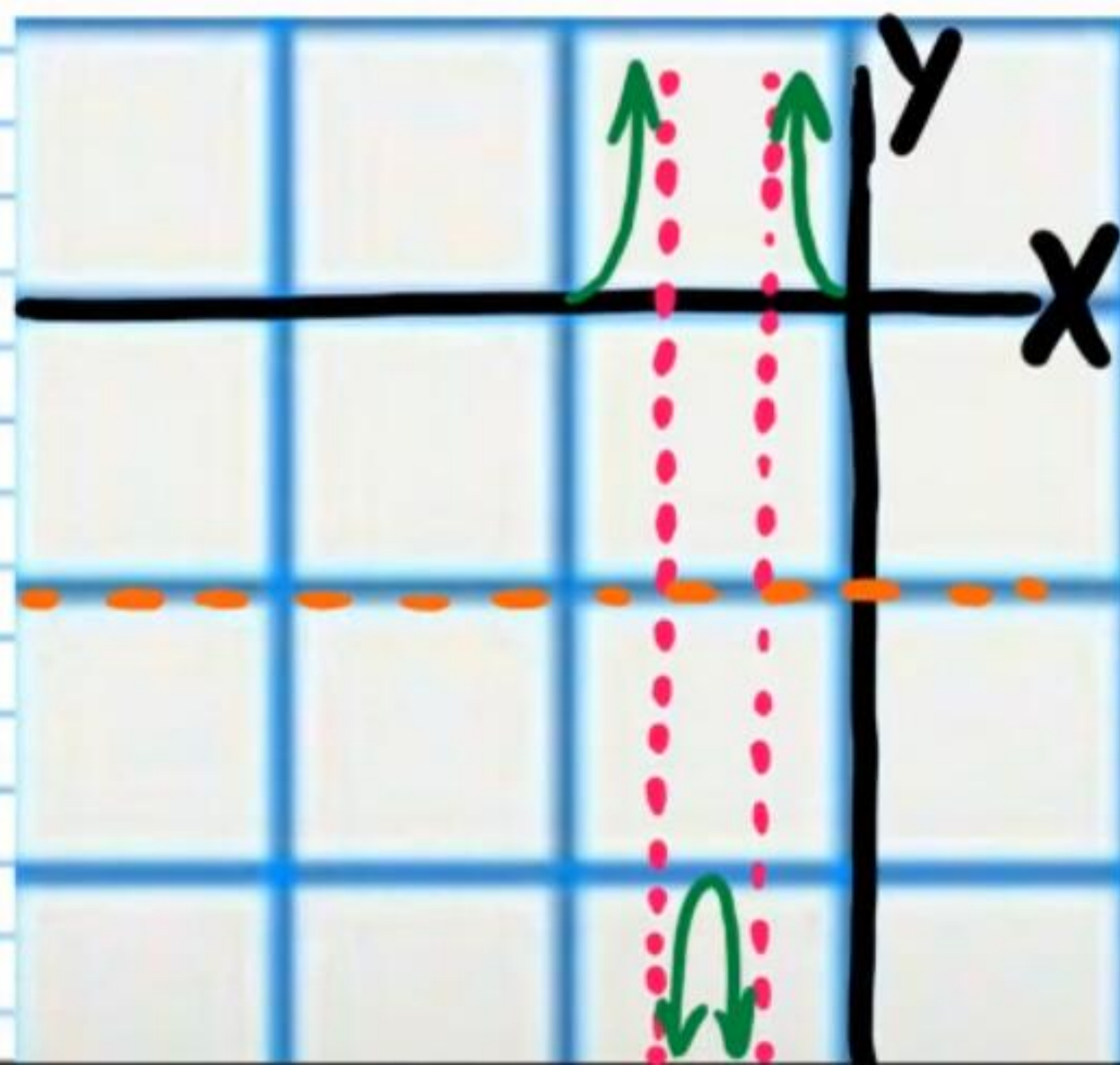
⑳ $\sin C \neq 1.48$

No Triangle Possible

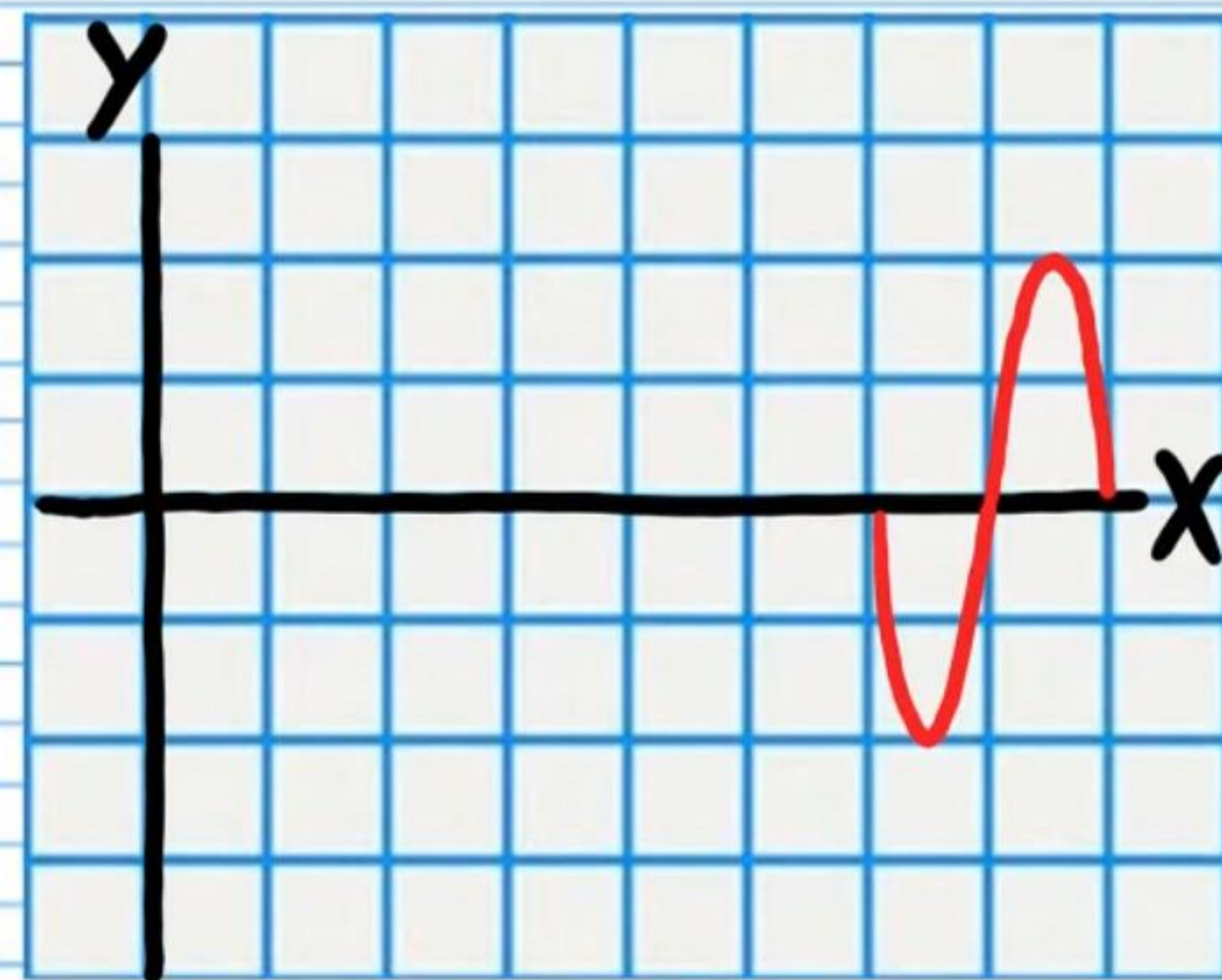
②① $B \approx 47.73^\circ$
 $A \approx 75.27^\circ$
 $a \approx 39.21 \text{ mi}$

②② $C_1 \approx 35.93^\circ$
 $B_1 \approx 116.07^\circ$
 $b_1 \approx 12.25 \text{ in}$
 $C_2 \approx 144.07^\circ$
 $B_2 \approx 7.93^\circ$
 $b_2 \approx 1.88 \text{ in}$

②③ $y = -1$
 $P = 90^\circ$
 $PS = -90^\circ$



②④ $y = 0$
 $P = 180^\circ$
 $PS = 540^\circ$
 $A = 2$



②⑤ $y = 1$
 $P = 720^\circ$
 $PS = -180^\circ$

